

OPTOELECTRONICS & DEVICE PHYSICS

Module 1

“Fundamentals of Materials”

CONTENT

- Concept of energy bands
- Charge carriers
- Carrier concentration
- Concept of Fermi level
- Hall effect
- Superconductors
- Numericals

Band Theory- Concept of energy bands

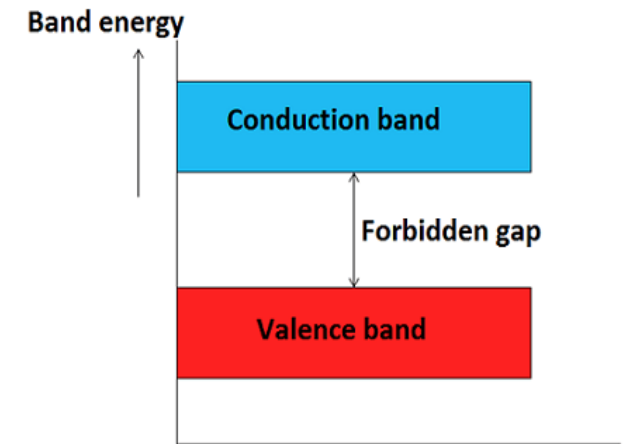
- The band theory of solids is a fundamental concept in condensed matter physics and materials science that helps explain the electronic structure and electrical conductivity of solid materials.
- It provides a framework for understanding how electrons are arranged and behave in a crystalline solid.
- It helps explain why some materials are insulators, some are semiconductors, and others are conductors, based on the energy band structure and the presence or absence of an energy band.

The **key ideas** and principles of band theory are:

1. Energy Bands: In a crystalline solid, the atoms are arranged in a periodic lattice structure. When electrons from individual atoms come together in a solid, they form energy bands. Energy bands represent the allowed energy levels that electrons can occupy in the crystal lattice.

2. Valence and Conduction Bands: The two most important energy bands are the valence band and the conduction band. The valence band is the lower-energy band, and it is typically filled with electrons. Electrons in the valence band are tightly bound to their respective atoms. The conduction band is the higher-energy band, and it is usually partially or completely empty. Electrons in the conduction band are free to move throughout the crystal.

3. Energy Band Gap: The region of energy levels between the valence band and the conduction band is called the energy band gap (or bandgap). It represents the energy range where electrons are forbidden to exist. The size of the band gap is a crucial factor in determining a material's electrical properties.



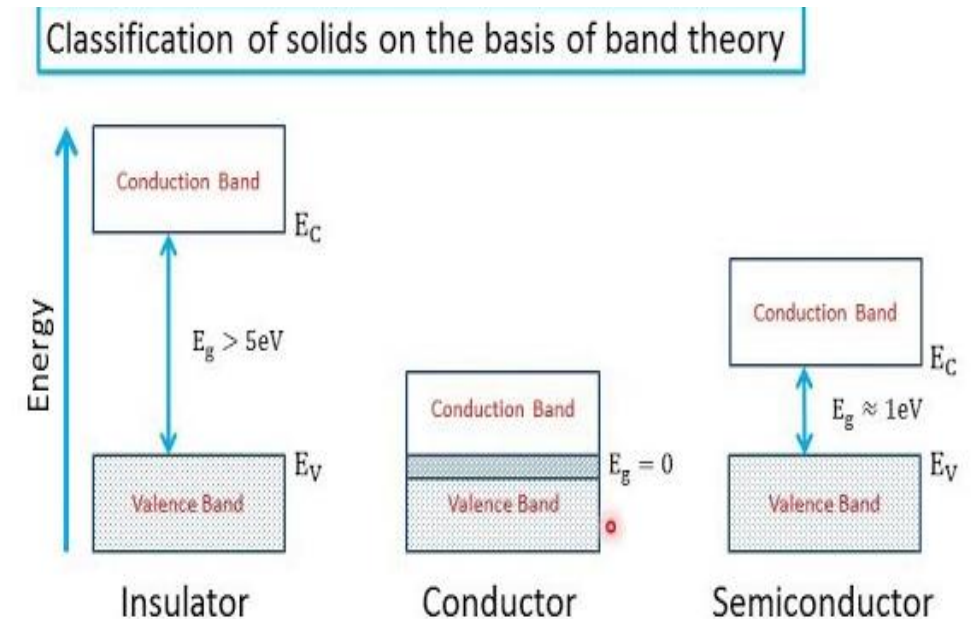
Classification of Solids on the Basis of Band Theory

Based on the energy band gap, materials can be classified into three main categories:

1. **Insulators:** Materials with a large band gap that have very few electrons in the conduction band, making them poor conductors of electricity are called insulators e.g., diamond, glass.

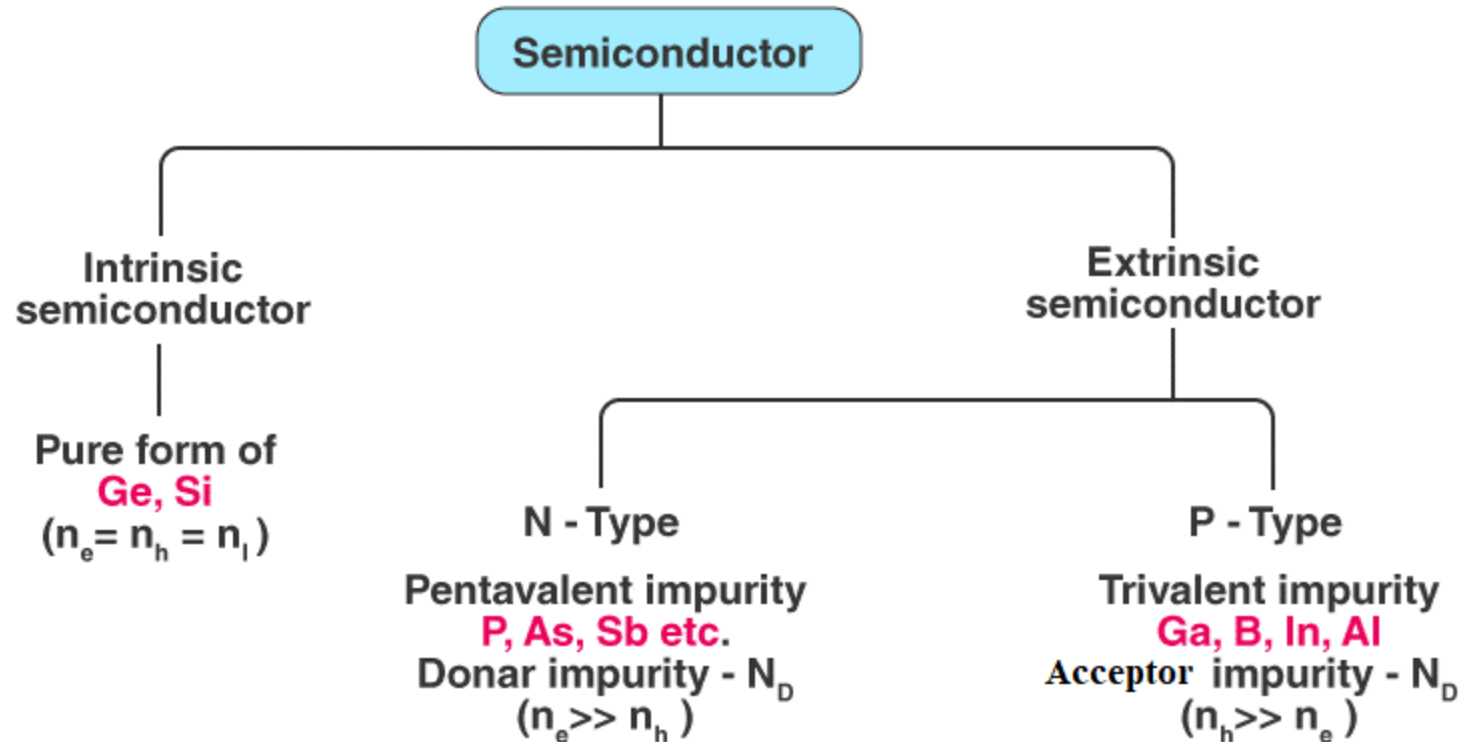
2. **Semiconductors:** Materials with a moderate band gap that have a few electrons in the conduction band at room temperature, and their electrical conductivity can be controlled by adding small amounts of impurities (doping) or by applying external factors like temperature are called semiconductors e.g., silicon, germanium .

3. **Conductors:** Materials with overlapping valence and conduction bands that have a large number of free electrons in the conduction band and are excellent conductors of electricity e.g., metals.



Classification of Semiconductors

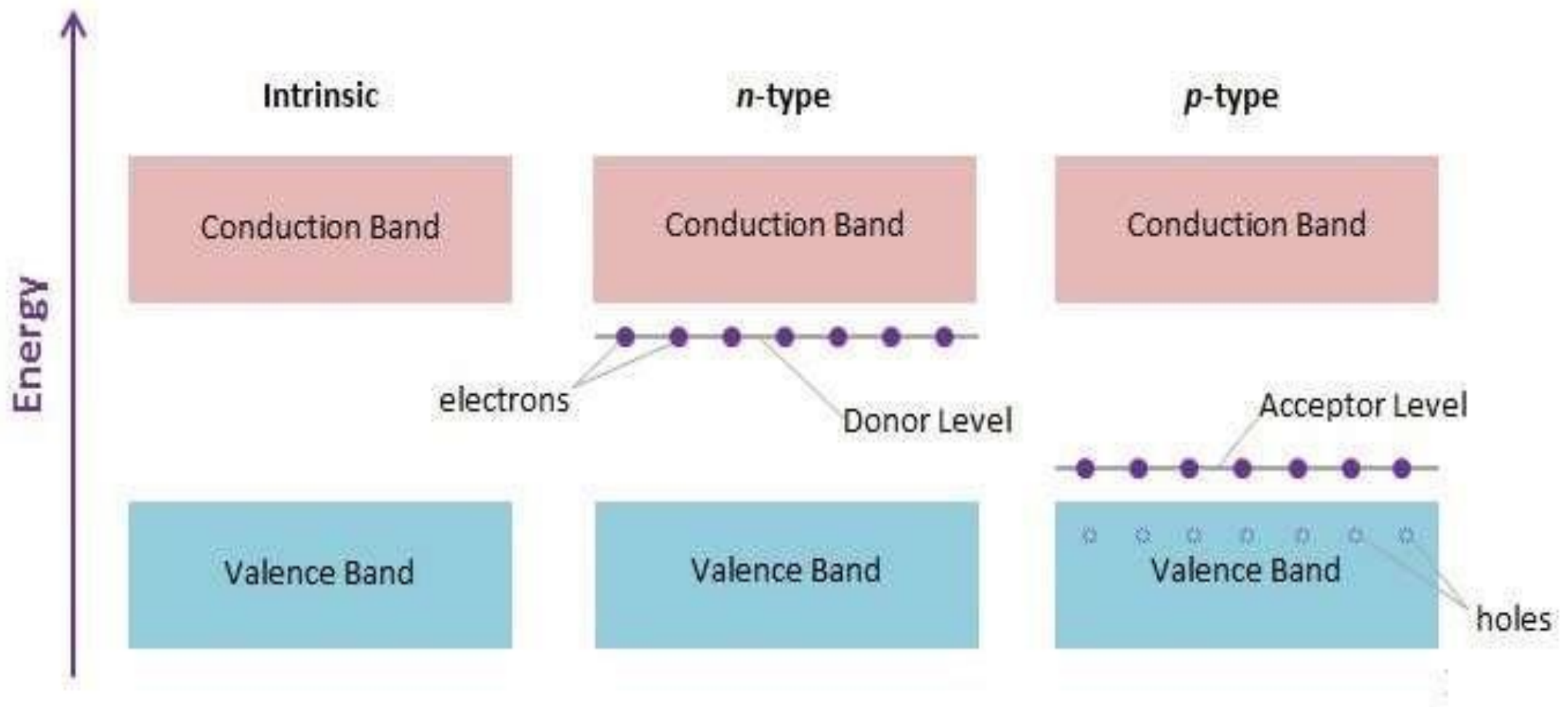
- Semiconductors are materials that have electrical conductivity between that of conductors (such as metals) and insulators (such as ceramics or plastics).
- They are essential components in electronic devices and can be classified in several ways



Doping: The process of adding controlled impurities to a semiconductor is known as doping

Differentiate N-type and P-type Semiconductors

Sl No	N- type semiconductor	P-type Semiconductor
1	Extrinsic semiconductor obtained by doping with pentavalent impurity such as Phosphorous, Antimony, Arsenic etc	Extrinsic semiconductor obtained by doping with trivalent impurity such as Boron, Gallium, Indium etc
2	The impurity atoms added provide extra electrons in the structure, and are called donor atoms.	The impurity atoms added create vacancies of electrons (i.e., holes) in the structure, and are called acceptor atoms.
3	Majority Charge carriers are electrons and minority charge carriers are holes.	Majority Charge carriers are holes and minority charge carriers are electrons.



Energy band diagram of intrinsic, *n*-type and *p*-type semiconductors

Difference Between Intrinsic and Extrinsic Semiconductors

S.NO	Intrinsic Semiconductor	Extrinsic Semiconductor
1	It is in pure form.	It is formed by adding trivalent or pentavalent impurity to a pure semiconductor.
2	Holes and electrons are equal.	No. of free holes are more in p-type and no. of free electrons are more in n-type.
3	Fermi level lies in between valence and conduction Bands.	Fermi level lies near valence band in p-type and near conduction band in n-type.
4	Ratio of majority and minority carriers is unity.	Ratio of majority and minority carriers is not unity.

Carrier concentration in an intrinsic semiconductor

- ❑ The number of electrons in the conduction band per unit volume and the number of holes in the valence band per unit volume of the material is called carrier concentration or density of charge carriers.
- ❑ The number of electrons in the conduction band per unit volume of the material is called the **electron concentration**.
- ❑ The number of holes in the valence band per unit volume of the material is called the **hole concentration**.
- ❑ **Expression for electron concentration in conduction band:**

$$n = 2 \left[\frac{2\pi m_e kT}{h^2} \right]^{3/2} e^{\frac{-(E_C - E_F)}{kT}}$$

where E_C is the energy of conduction band and E_F – fermi energy

- ❑ **Expression for hole concentration in valence band:**

$$p = 2 \left[\frac{2\pi m_h kT}{h^2} \right]^{3/2} e^{\frac{-(E_F - E_V)}{kT}}$$

❑ Expression for intrinsic carrier concentration:

In the case of intrinsic semiconductors $n = p$

$$\text{Therefore } n_i = 2 \left[\frac{2\pi kT}{h^2} \right]^{3/2} (m_e m_h)^{3/4} e^{-\frac{(E_g)}{2kT}}$$

❑ Expression for fraction of electron in conduction band in an intrinsic semiconductor:

$$\frac{n}{N} = e^{-\left(\frac{E_g}{2kT}\right)}$$

where n is the number of electrons excited to conduction band N is the total number of electrons available in the valence band initially.

1. Compute the concentration of intrinsic charge carriers in a germanium crystal at 300 K. Given that $E_g = 0.72$ eV and assume that $m_e = m_h$.

Ans

$$n_i = 2 \left[\frac{2\pi kT}{h^2} \right]^{3/2} (m_e m_h)^{3/4} e^{-\frac{E_g}{2kT}}$$

Answer is $2.27 \times 10^{19} / \text{m}^3$

2. Estimate the fraction of electrons in the conduction band at 300K of

- (a) Germanium ($E_g = 0.72 \text{ eV}$)
- (b) Silicon ($E_g = 1.1 \text{ eV}$) and
- (c) Diamond ($E_g = 5.6 \text{ eV}$).
- (d) What is the significance of these results?

Ans

$$\frac{n}{N} = e^{-\left(\frac{E_g}{2kT}\right)}$$

a) Germanium: fraction = **9.07×10^{-7}**

b) Silicon: fraction = **5.86×10^{-10}**

c) Diamond: fraction = **1.008×10^{-47}**

d) The results shows that, larger the bandgap smaller the electrons that can go into the conduction band, at a given temperature.

3. Estimate the fraction of electrons in the conduction band at room temperature in Ge with $E_g=0.72$ eV.

Ans

$$\frac{n}{N} = e^{-\left(\frac{E_g}{2kT}\right)}$$

a) Germanium: fraction = **9.07×10^{-7}**

4. Assuming that the number of electrons near the top of the valence band available for conduction is $5 \times 10^{25} /m^3$ and the intrinsic carrier density is $2.5 \times 10^{19} /m^3$, calculate the energy gap of germanium at room temperature.

$$\frac{n}{N} = e^{-\left(\frac{E_g}{2kT}\right)}$$

$$E_g = -2kT \ln\left(\frac{n}{N}\right)$$

$$E_g = -2kT \ln\left(\frac{5 \times 10^{25}}{2.5 \times 10^{19}}\right)$$

$$E_g = 2kT(14.508)$$
$$E_g = \frac{1.201 \times 10^{-19}}{1.6 \times 10^{-19}}$$

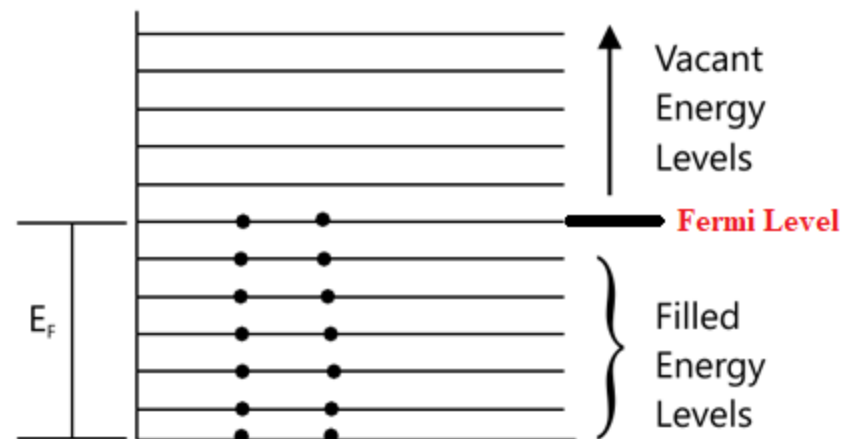
Ans is 0.75 eV

Fermi Energy

- Fermi energy, named after the Italian physicist Enrico Fermi.
- It is a fundamental concept in quantum mechanics and solid-state physics.
- It represents the highest energy state of an electron at absolute zero temperature (0 Kelvin or -273.15 degrees Celsius) in a system.

Definition

- The energy corresponding to the highest occupied level of electrons at 0 K is called **Fermi energy**.
- The corresponding energy level is called the **Fermi level**.
- At 0 K all the energy levels above Fermi level are **empty** and all those below are **completely filled**.



Key points about Fermi energy:

1. Quantum Mechanical Basis: In a solid or any other system of fermions (particles with half-integer spin, such as electrons), the **Pauli exclusion principle** dictates that no two electrons can occupy the same quantum state. This principle results in a distribution of electron energies called the **Fermi-Dirac distribution**.

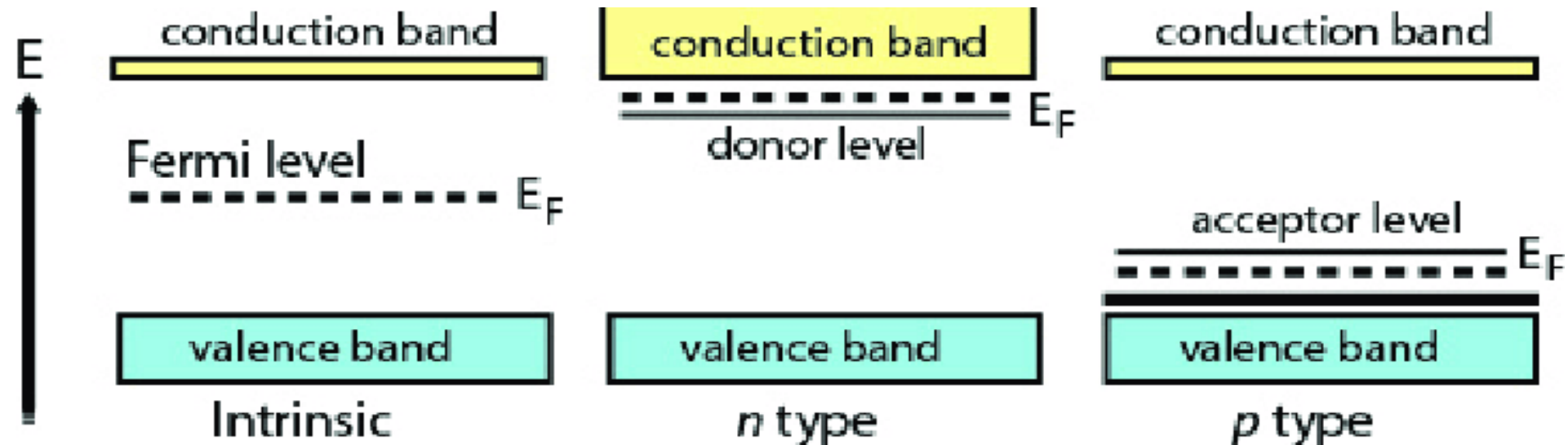
2. Absolute Zero Temperature: Fermi energy is defined at absolute zero temperature because at this temperature, all available electron states up to the Fermi energy level are occupied, while all states above it are unoccupied.

3. Role in Conductivity: Fermi energy is a crucial factor in determining the electrical conductivity of a material. Materials with a high Fermi energy have a greater number of electrons near the top of their energy bands, making them good conductors. Conversely, materials with a low Fermi energy have fewer electrons near the top of their bands and are typically insulators.

Key points about Fermi energy Continues:

4. **Temperature Effects:** At temperatures above absolute zero, electrons can gain thermal energy and occupy states above the Fermi energy, contributing to electrical conductivity. The probability distribution of electron energies broadens with increasing temperature.

5. **Application in Semiconductor Physics:** Fermi energy plays a crucial role in semiconductor physics, where it helps explain the behavior of charge carriers (electrons and holes) and their conductivity properties. It is a key parameter in the analysis of semiconductors.



Hall Effect

- ✓ Hall Effect was discovered by Edwin Herbert Hall in 1879.

Definition:

When a current-carrying conductor or a semiconductor is introduced to a perpendicular magnetic field, a voltage can be measured at the right angle to the current path (a transverse electric field is developed). This effect of obtaining a measurable voltage is known as the Hall Effect and the voltage is called Hall voltage.

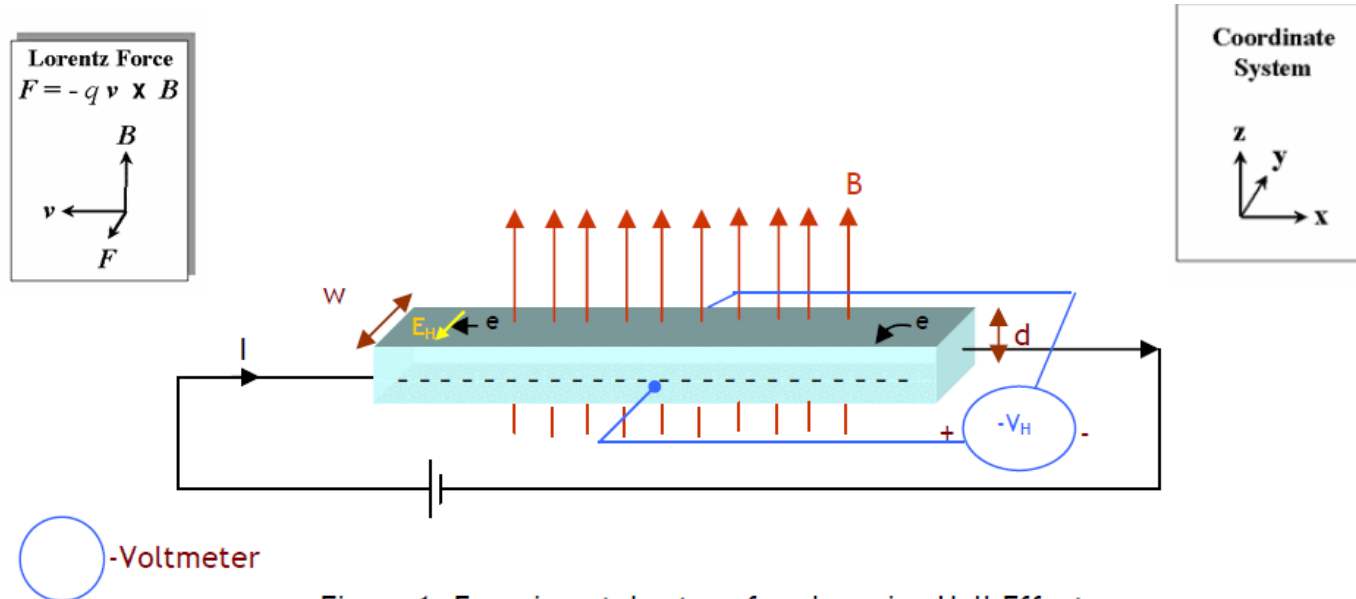
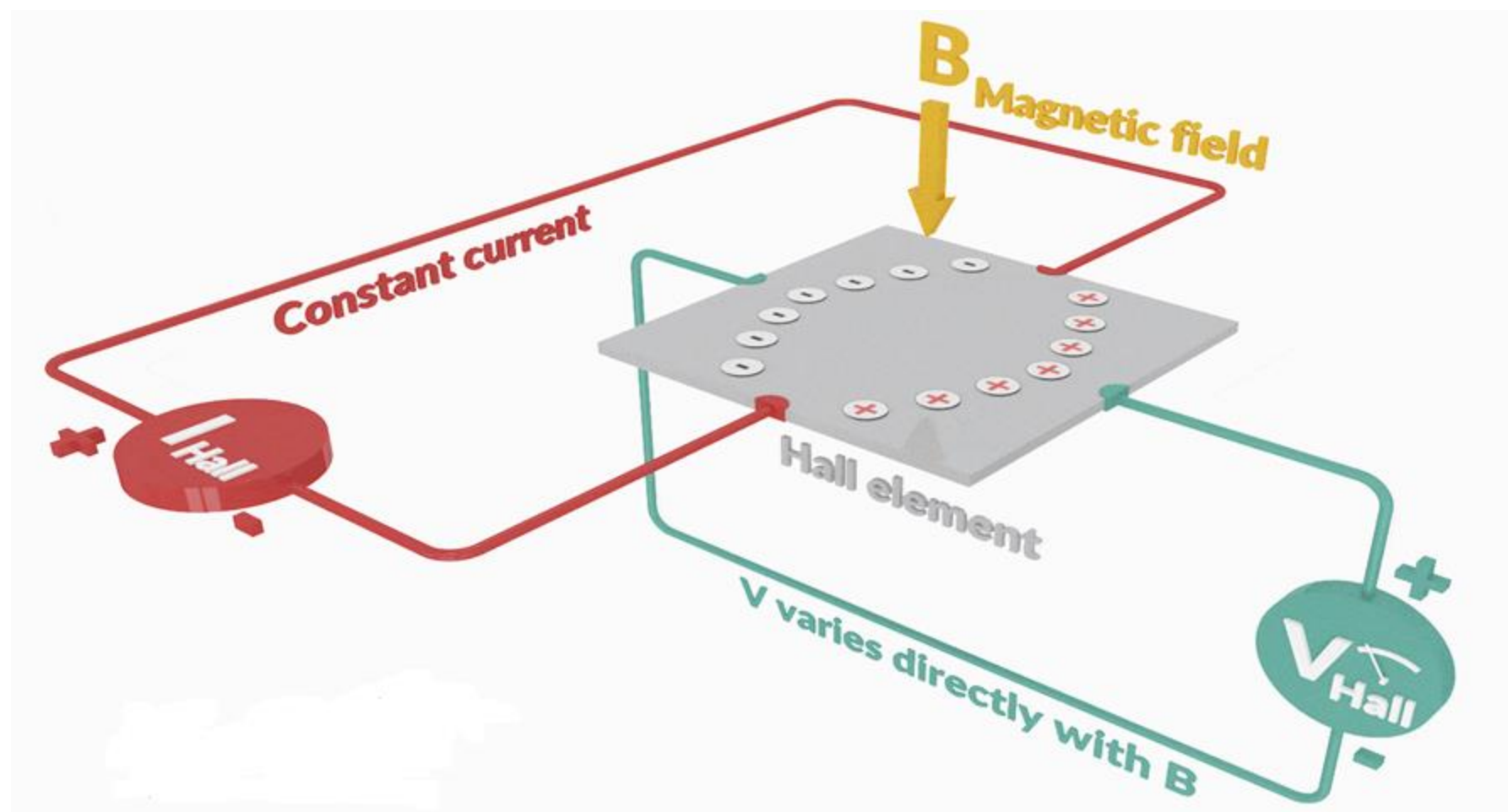


Figure 1: Experimental set-up for observing Hall Effect

Theory

When a conductive plate is connected to a circuit with a battery, then a current starts flowing. The charge carriers will follow a linear path from one end of the plate to the other end. The motion of charge carriers results in the production of magnetic fields. When a magnet is placed near the plate, the magnetic field of the charge carriers is distorted. This upsets the straight flow of the charge carriers. The force which disturbs the direction of flow of charge carriers is known as [Lorentz force](#).

Due to the distortion in the magnetic field of the charge carriers, the negatively charged electrons will be deflected to one side of the plate and positively charged holes to the other side. A potential difference, known as the Hall voltage will be generated between both sides of the plate which can be measured using a meter.



The Hall voltage

represented as V_H is given by the formula:

$$V_H = \frac{IB}{qnd}$$

Here,

I is the current flowing through the sensor

B is the magnetic Field Strength

q is the charge

n is the number of charge carriers per unit volume

d is the thickness of the sensor.

Hall Coefficient

The Hall coefficient R_H is mathematically expressed as

$$R_H = \frac{E}{JB}$$

Where, J is the current density of the carrier electron,
E is the induced electric field and B is the magnetic strength.

- The hall coefficient is positive if the number of positive charges is more than the negative charges.
- Similarly, it is negative when electrons are more than holes.

Other formulae

- $R_H = \frac{-1}{ne}$ or $R_H = \frac{1}{pe}$
- $R_H = \frac{V_H d}{BI}$

1. Calculate the Hall voltage when a conductor carrying a current of 100 A, is placed in a magnetic field of 1.5 T. The conductor has a thickness of 1 cm, and the number density of charges inside the conductor is $5.9 \times 10^{28} / \text{m}^3$.

$$V_H = \frac{IB}{qnd}$$

$$\text{Ans} = 15.89 \times 10^{-7} \text{ V}$$

2. The Hall coefficient of certain silicon specimen was found to be $-7.35 \times 10^{-5} \text{ m}^3 \text{ C}^{-1}$ from 100 to 400 K. Calculate the number of charge carriers and determine the nature of the semiconductor.

$$\text{Hall co-efficient } R_H = -7.35 \times 10^{-5} \text{ m}^3 \text{ C}^{-1}$$

a) *Density of electrons*

$$n = \frac{-1}{R_H e} \text{ (from equation (1))}$$

$$n = \frac{1}{(7.35 \times 10^{-5} \times 1.609 \times 10^{-19})}$$

$$\text{(i.e)} = 8.455 \times 10^{22} \text{ m}^{-3}$$

Nature is n type semiconductor.

3. A semiconducting crystal with 12 mm long, 5 mm wide and 1 mm thick has a magnetic density of 0.5 Wbm^{-2} applied from front to back perpendicular to largest faces. When a current of 20 mA flows lengthwise through the specimen, the voltage measured across its width is found to be $37\mu\text{V}$. What is the Hall coefficient of this semiconductor?

$$R_H = \frac{V_H d}{BI}$$

$$\text{Hall coefficient } R_H = 3.7 \times 10^{-6} \text{ C}^{-1} \text{ m}^3$$

Applications of Hall Effect

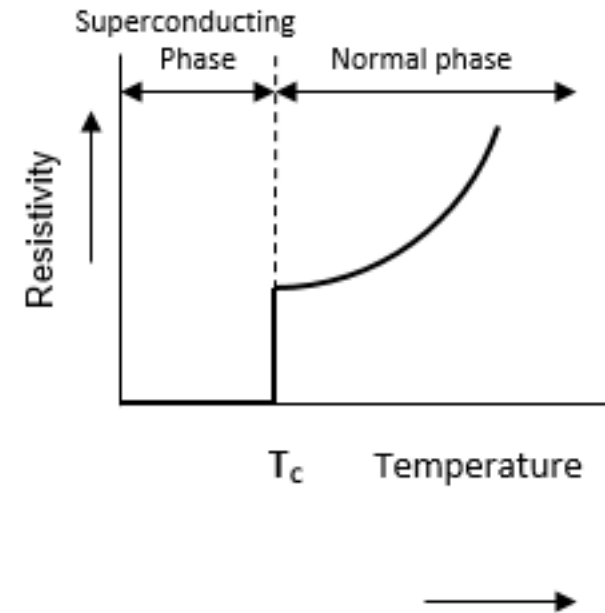
- Find whether the semiconductor is N-type or P-type
- Find carrier concentration
- Magnetic field sensing equipment
- Proximity detectors
- Hall effect Sensors and Probes

Superconductivity

- Superconductivity is a state of materials in which their **electrical resistance is zero**.
- The resistivity of metals increases linearly with increase in temperature.
- For certain metals, when they are cooled, the resistivity abruptly falls to zero at a temperature characteristic of the metal. This phenomenon is known as **superconductivity**.
- The temperature at which **transition takes place** from normal conducting to superconducting state is the transition temperature or critical temperature T_C .
- The phenomenon was discovered by Kamerlingh Onnes during his studies on the properties of mercury at very low temperatures.
- He found that the resistivity of mercury suddenly dropped to zero at **4.2 K**.

Temperature Dependence of Resistivity of Superconductors

- The variation of resistivity ' ρ ' of a superconductor with temperature is shown in the graph.
- As temperature is decreased the resistivity decreases and suddenly falls to zero at T_C , which is called the critical temperature.
- Below T_C it continues to remain a superconductor.
- Critical temperature is different for different materials.
- Mercury loses resistance completely at 4.2 K, whereas above this temperature mercury behaves as a normal conductor.



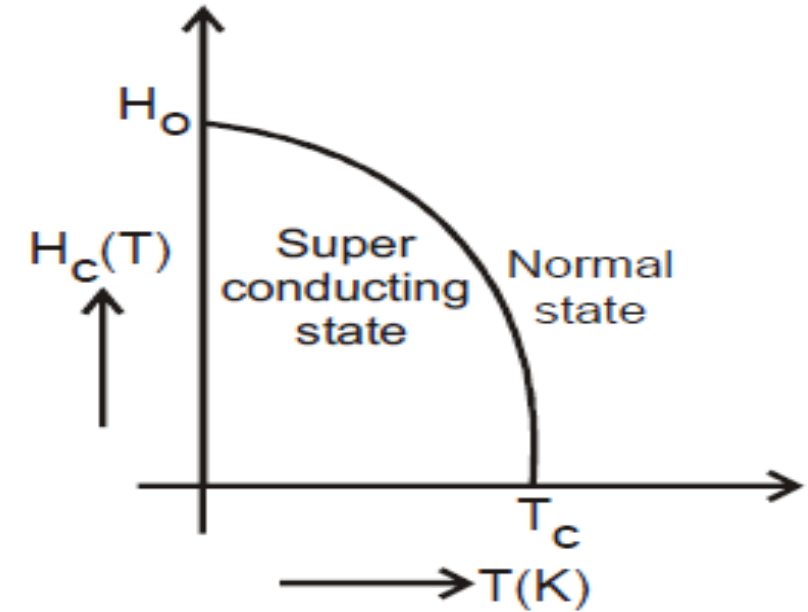
General features (characteristics) of superconductors

1. Critical field H_C

- Superconductivity vanishes when a sufficiently strong magnetic field is applied.
- “The minimum magnetic field required to destroy the superconductivity in the material is called **critical field H_C** ”.
- Critical field is a function of temperature and varies with temperature T in accordance with the relation

$$H_C = H_0 \left(1 - \left(\frac{T}{T_C} \right)^2 \right)$$

where H_0 is the critical magnetic field at 0 K.



Variation of critical field as a function of temperature

1. A superconducting tin has a critical temperature of 3.7 K at zero magnetic field and a critical field of 0.0306 Tesla at 0 K. Find the critical field at 2 K.

Solution:

Given data:

Critical temperature $T_c = 3.7 \text{ K}$

Critical field $H_c = 0.0306 \text{ Tesla}$

Temperature $T = 2 \text{ K}$

The critical magnetic field

$$\begin{aligned} H_c &= H_0 \left[1 - \left[\frac{T^2}{T_c^2} \right] \right] \\ &= 0.0306 \left[1 - \left[\frac{2}{3.7} \right]^2 \right] = 0.0216 \text{ Tesla} \end{aligned}$$

The critical magnetic field $H_c = 0.0216 \text{ Tesla}$.

2. The critical temperature of Nb is 9.15K. At zero kelvin the critical field is 0.196 T. In the laboratory the temperature of Nb is increased

i) Using a magnetic field, Nb can be changed to normal conductor below critical temperature. State **True** or False.

ii) The magnetic field required to change superconducting Nb to normal conductor at 0K is

a) 0 T

b) 1 T

c) 0.5 T

d) **0.196 T**

iii) The magnetic field required to change superconducting Nb to normal conductor above 0K and below 9.15K is

a) More than 0.196 T

b) **Less than 0.196 T**

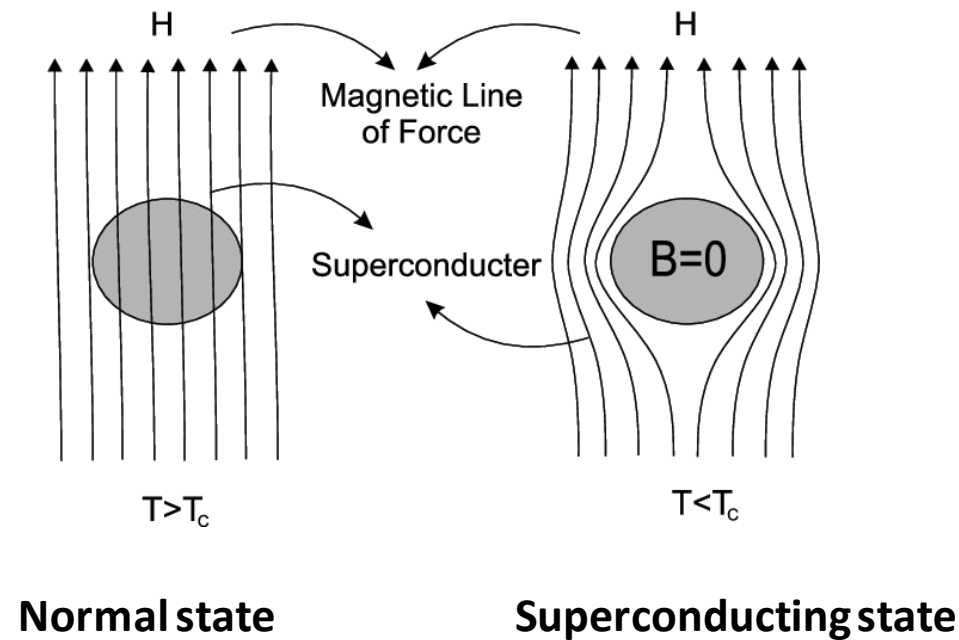
c) Equal to 0.196T

d) Zero

2. Meissner effect

When a specimen of a conducting material is placed in a magnetic field and cooled below its critical temperature, the magnetic field is completely expelled from the specimen.

“The phenomenon of expulsion of magnetic flux completely from the specimen during the transition from normal state to the superconducting state is called as ‘Meissner effect.’”



The magnetic induction inside the specimen is given by

$$\mathbf{B} = \mu_0 (\mathbf{H} + \mathbf{M})$$

Where, B is magnetic induction,

H is external applied field and

M is magnetization produced within the specimen.

At $T < T_c$, $\mathbf{B} = 0$ inside the super conductor.

Hence we can write,

$$\mathbf{B} = \mu_0 (\mathbf{H} + \mathbf{M}) = 0,$$

$$\text{or } \mathbf{H} = -\mathbf{M}$$

$$M/H = -1$$

$\chi = -1$ (Negative) , which is the characteristic of diamagnetism

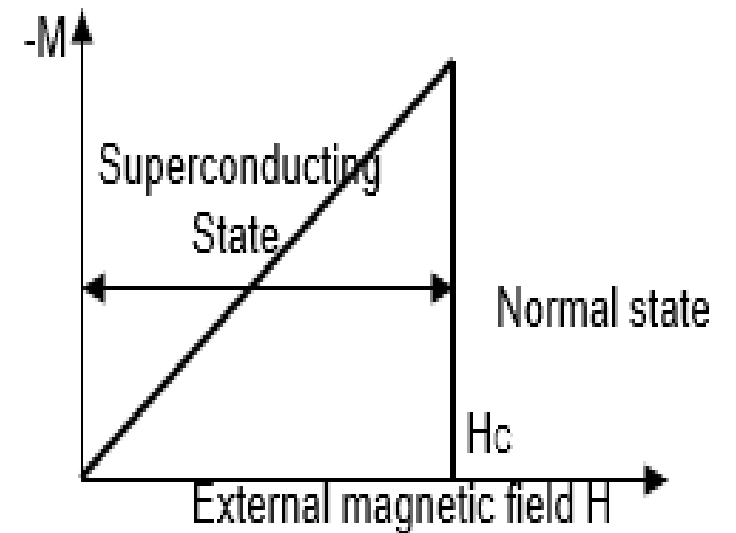
This shows that the superconductors are perfectly **diamagnetic in nature**.

Types of Superconductors

Based on the magnetic behavior, superconductors are classified into two categories, viz. Type I and Type II superconductors.

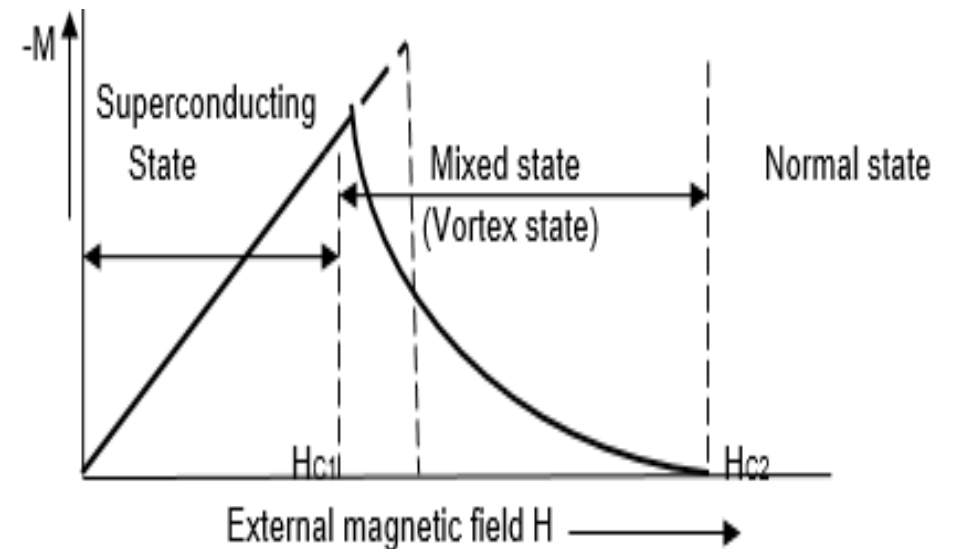
Type I Superconductors (Soft Superconductors):

- In type I superconductors, the transition from superconducting to normal state, in the presence of magnetic field, occurs sharply at the critical field H_C .
- Below the critical field the magnetization in the specimen is proportional to the applied field and it behaves like a diamagnet.
- For $H > H_C$, magnetization of the specimen is negligible and the specimen becomes a normal conductor.
- This kind of behavior is exhibited by pure elements. The critical field is relatively low for these superconductors.
- Examples are aluminium, lead and indium.



Type II Superconductors (Hard Superconductors):

- Type II superconductors are characterized by two critical fields H_{C1} and H_{C2} .
- Up to H_{C1} , below the critical field, the specimen is perfectly diamagnetic and hence is superconducting.
- From H_{C1} to H_{C2} , Meissner effect is incomplete but the specimen exhibits superconductivity.
- This state is called the **mixed or vortex state**.
- Above the critical field H_{C2} , the specimen becomes an ordinary conductor.
- H_{C2} can be as high as 30 T, and retention of superconductivity at such high magnetic fields makes them useful in creating very high magnetic fields.
- Examples of type II superconductors are transition metals and alloys containing niobium, silicon and vanadium

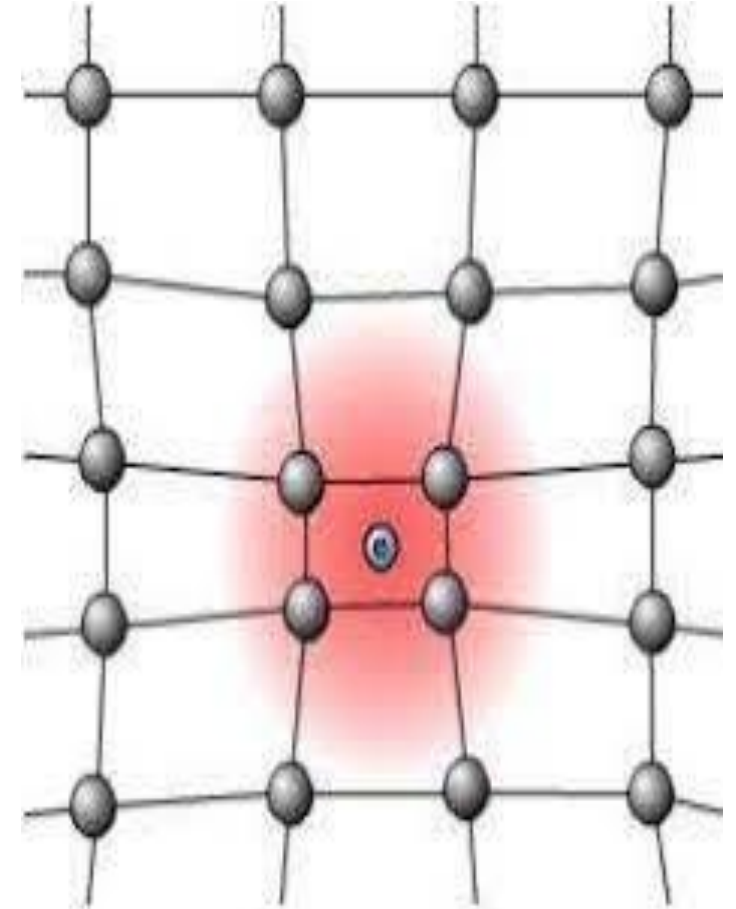


Type-I and Type-II superconductors

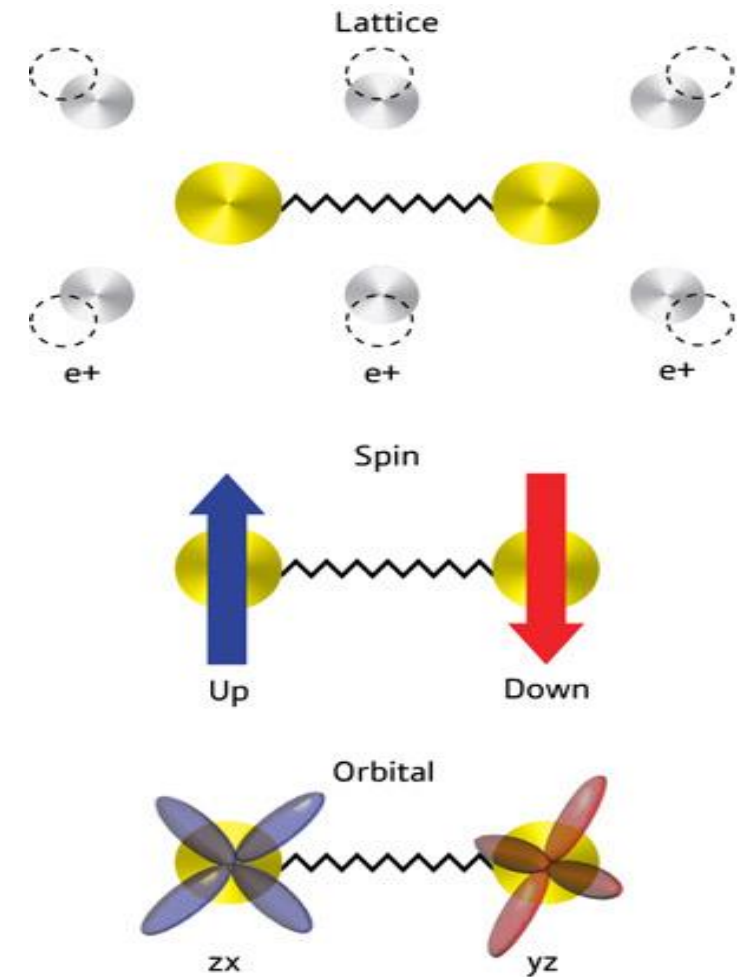
S.No	Type-I superconductors	Type-II superconductors
1	The material loses its magnetization abruptly.	The material loses its magnetization gradually.
2	They exhibit complete Meissner effect	They do not exhibit complete Meissner effect
3	They are characterized by one critical magnetic field (H_c)	They are characterized by two critical magnetic fields lower critical field (H_{c1}) and upper critical field (H_{c2})
4	No mixed state is present.	Mixed state is present
5	They are called soft superconductors. ($H_c=0.1T$)	They are called Hard superconductors ($H_{c2}=30T$)
6	Ex: Lead, Tin, Hg, Al etc.	Ex: Nb-Sn, Nb-Zr, Nb-Ti etc.

BCS Theory

- Explanation to superconductivity was given by John Bardeen, Leon N Cooper and John Robert Schrieffer on the basis of quantum theory. The theory is named after the three scientists as BCS theory.
- When an electron approaches a positive ion, there is a Coulomb attraction between them, which produces a distortion in the lattice.
- Now if another electron passes through the distorted lattice, interaction between the distorted lattice and the electron occurs, which results in lowering of energy of the electron.
- This effect is an interaction between two electrons via the lattice distortion or the phonon (quantum of energy of lattice vibration), which results in the lowering of energy for the two electrons.



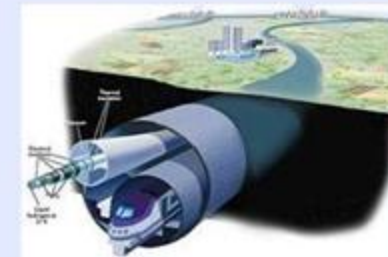
- This lowering of energy implies that the force between the two electrons is attractive. This type of interaction is called the electron-phonon-electron interaction.
- This interaction is strongest if the two electrons **have equal and opposite momentums and spin.**
- Such pairs of electrons are called **Cooper pairs.**
- At very low temperatures the density of Cooper pairs is large and they collectively move through the lattice with small velocity.
- **Low speed reduces collisions** and thereby resistivity decreases, which explains superconductivity.
- The attraction between the electrons in the Cooper pair is very weak. Therefore they can be separated by small increase in temperature, which results in a transition back to the normal state.



Applications

Applications using Superconductors

MRI Body scanners
Accelerator magnets
Magnetometers for
medicine,
seismography ...
Transport
Power transmission



Sample Questions: Remembering

1. Phosphorous can be used as
 - a) **Pentavalent Impurity**
 - b) Trivalent Impurity
 - c) Intrinsic semiconductor
 - d) Extrinsic Semiconductor
2. The temperature at which conductivity of a material becomes infinite is called
 - a) **Critical Temperature**
 - b) Absolute Temperature
 - c) Mean Temperature
 - d) Crystallization temperature
3. Superconductors are paramagnetic in nature. State True/ **False**

Sample Questions :Thought Provoking

1. Fermi energy of a Copper wire is found to be 8.5 eV. If the length of the wire is doubled, the Fermi energy will be

- a) **8.5 eV**
- b) 4.25 eV
- c) 17 eV
- d) 65 eV

2. For calculating the Fermi energy of a material, the graph of Resistance vs. Temperature is plotted. The slope is found to be exactly same as the slope obtained for Copper. The density of Copper is 8.96 g/cm³ and the density of the material is 10 g/cm³. We know that Fermi Energy is proportional to the square root of the product of density and the slope of the above graph, ($E_F \propto \sqrt{\rho S}$). If everything else remains same, we can say for sure that the Fermi energy of the material is

- a) lower than that of copper
- b) same as that of copper
- c) **higher than that of copper**
- d) inverse of the Fermi energy of copper

Thank You